

# ISAAC (ZAC) ADJEI

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## PROFILE

Electronic Engineering and Computer Science student at Aston University (Predicted First Class) with strong security mindset. Implements secure authentication, audit logging and RBAC. Seeking cybersecurity internships.

## EDUCATION

### Aston University, Birmingham, United Kingdom

Oct 2024 - Jul 2028

BEng (Hons), Electronic Engineering and Computer Science · **Predicted: First Class**

Top 40 Finalist - Black Heritage Undergraduate of the Year Award 2026 (TargetJobs & Sky)

**Relevant Modules:** Internet Applications and Databases, Foundations of AI and Data Science, Introductory Programming for Digital Science (Python), Electronics 1, Electronics 2, Introductory Mathematics for Engineering, Power Skills

- Developing strong foundations in software engineering, electronics and applied mathematics while building practical project experience across embedded systems, AI/data and full-stack development

### Stanmore College, London, United Kingdom

Sep 2022 - Jul 2024

Pearson BTEC Level 3 National Extended Diploma in Engineering · **D\*DD (Distinction\*, Distinction, Distinction)**

Best and Most Hardworking Student Award

**Relevant Modules:** Engineering Product Design and Manufacture, Microcontroller Systems for Engineers, Engineering Principles, Specialist Engineering Project, CAD in Engineering, Electronic Devices and Circuits, Calculus to Solve Engineering Problems, Electronic Measurement and Testing of Circuits

- Completed intensive hands-on training in engineering design, testing and documentation, applying microcontroller, electronics and CAD skills through practical coursework and projects

## SKILLS

**Cloud & DevOps:** AWS, Azure, Docker, Kubernetes, Vercel, Cloudflare, GitHub Actions

**Languages:** C, C++, Python, Java, TypeScript, JavaScript, PHP, SQL, MATLAB

**Web and Frameworks:** Next.js, React, Node.js, Tailwind CSS, PHP, MySQL, Apache

**Embedded & Hardware:** Arduino, STM32, ESP32, Raspberry Pi, AVR, KiCad, Proteus, PCB Layout, Circuit Design, Oscilloscope, I2C / UART / SPI, FPGA, VHDL

**AI/ML and Data:** TensorFlow, PyTorch, NumPy, Pandas, Scikit-Learn, OpenCV, Jupyter Notebooks

**Professional:** Problem Solving, Critical Thinking, Communication, Team Collaboration, Leadership, Time Management, Adaptability, Attention to Detail

## PROJECTS

### AstonCV - Full-Stack CV Database

2026

PHP, MySQL, CSS, Apache, Cloudflare

[Website ↗](#) | [GitHub ↗](#) | [Portfolio ↗](#)

- Engineered **12+ security controls** including **bcrypt hashing**, **CSRF tokens**, **PDO prepared statements** and **brute-force protection**
- **Deployed live** with searchable profiles, sorting/filtering and server-side PDF export

### PHAEMOS: Smart Maintenance Platform

2026, ongoing

C++ (ESP32, Arduino), C (STM32 HAL), MicroPython (Pico 2W), FastAPI, Next.js 15,

[Website ↗](#) | [GitHub ↗](#) | [Portfolio ↗](#)

PostgreSQL 15, Redis 7, scikit-learn, Docker, Prometheus, Grafana

- Architected a **4-node IoT platform** (ESP32, STM32 Black Pill, Arduino Nano, Pico 2W/MicroPython) collecting **18 real-time sensor streams** (vibration, IR, gas, RPM, moisture, sound); **FastAPI + PostgreSQL 15 + Redis 7** backend processing telemetry in under **200ms per request**
- **Isolation Forest anomaly detection** on every ingest, auto-generating maintenance tickets with diagnostic recommendations; STM32 runs **100Hz FFT vibration analysis** via ARM CMSIS-DSP (**9x speedup** over naive DFT)
- **JWT refresh flow**, **Google/GitHub OAuth**, **TOTP 2FA**, multi-tenant **RBAC**, rate limiting, immutable audit log; **WebSocket** live dashboard; **Prometheus + Grafana** monitoring; deployed on **DigitalOcean VPS + Nginx**

### git-unlocked - Open Source Git Course

2026

Git, GitHub, GitLab, Markdown, Open Source

[GitHub ↗](#) | [Portfolio ↗](#)

- Created a structured learning resource with **217+ topic files** covering fundamentals to advanced workflows across Git, GitHub, GitLab and 5 other platforms
- Maintained contributor workflow with issue templates, PR standards and CI checks

### 4x4x4 NeoPixel LED Cube

2025

Arduino, C++, WS2812B, Embedded Systems

[GitHub ↗](#) | [Portfolio ↗](#)

- Built a 64-LED interactive display with a **non-blocking state machine architecture** using millis()
- Introduced four animation modes, ambient-light adaptive brightness and debounced user controls

## Two-Stage Audio Amplifier

Analogue Design, PCB Design, KiCad, Proteus, Electronics

2026

[Portfolio ↗](#)

- Delivered a **full engineering workflow: simulation, breadboard validation, PCB layout and bench testing**
- Validated performance with measured output of **2.980 Vpp** against a **3 Vpp** design target

## CNC Milling Machine Control System

Arduino, C++, Safety Systems, State Machines

2024

[Portfolio ↗](#)

- Engineered a **safety-focused controller** using **finite state machine logic** and strict transition validation
- Integrated door interlocks, **emergency stop handling**, timed cooldown and LCD status feedback for safe operation

## EXPERIENCE

### Student Representative

Student advocacy, stakeholder communication, academic liaison

**Aston Students' Union, Birmingham, UK**

Sep 2025 - Present

- Represent student interests through **SSLC, Senate and Council meetings**, ensuring feedback is accurately captured and addressed through formal academic governance channels
- Contributed to **academic governance and course quality** improvements through structured reporting and consistent follow-up on agreed actions

### Diplomatic, Consular and Estates Intern

Consular operations, diplomatic property, document processing, client support

**Ghana High Commission, London, UK**

Mar 2024 - Aug 2024

- Operated secure consular processing systems for identity verification, **biometric data capture**, document authentication and controlled visa and passport production at **100+ daily interactions**
- Maintained sensitive estates documentation and enforced secure record-keeping across diplomatic administrative workflows

### Junior Apprentice HVAC Technician

Field servicing, diagnostics, installation support

**Massive Refrigeration Services, Accra, Ghana**

Jul 2019 - Jul 2021

- Serviced, installed and maintained **50+ air conditioning units** using standard diagnostic tools and field servicing techniques

## RESEARCH & PUBLICATIONS

### git-unlocked: An Open-Source Modular Curriculum for Git and Version Control Education

I. Adjei · Zenodo · Jun 2026

[Zenodo ↗](#) | [DOI ↗](#)

- Open-source modular curriculum covering **12 thematic sections and 217+ topic files** across Git, GitHub and GitLab; published as a citable research artefact with a permanent DOI

## VOLUNTEERING

### Student Judge - National Emerging Talent Awards 2026

targetjobs UK (Group GTI)

Feb 2026

- Evaluated employer submissions for **Best Placement or Internship Programme** using qualitative and scoring criteria; commended for a thorough, timely and high-quality approach

### Fundraising Volunteer - 10 Days of 5K Challenge

Cancer Research UK

Feb - Mar 2026

- Completed **10 × 5km runs (50km+)** for the Cancer Research UK 10 Days of 5K Challenge, managing a fundraising page that contributed to a campaign raising over **£797,424.64** (plus £159,224.14 Gift Aid)

## SPOKEN LANGUAGES

**English** (Full professional), **Twi & Ga** (Full Native), **French & Spanish** (Elementary)